



ABSTRACT

Objective: We develop a repeatable decision framework to evaluate whether an OEM should enter a GT racing series by converting a production vehicle.

Approach: Using synthetic datasets, we construct a multi-stage gate model assessing platform feasibility, regulatory risk, customer economics, and brand impact.

Key Output: A structured scorecard generating Go / Conditional Go / No-Go recommendations across GT series tiers.

Impact: Enables OEMs to make transparent, data-driven motorsport investment decisions while balancing cost, brand exposure, and competitive positioning.

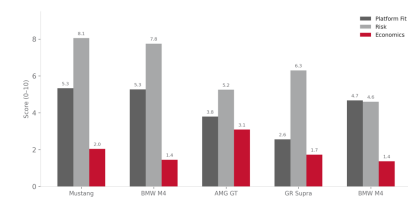
INTRODUCTION

Context: Private-tier GT racing offers OEMs opportunities in brand exposure, revenue generation, and downstream parts demand.

Challenge: Each series differs significantly in technical requirements, homologation complexity, and cost structure, making direct comparison difficult.

Problem: OEMs cannot pursue all opportunities and require a systematic way to prioritize viable entry options.

Core Question: Which GT series and platform combination provides the strongest Go / No-Go case based on market attractiveness, engineering feasibility, cost, and competitive dynamics?



METHODS AND MATERIALS

Framework: We develop a 6-stage gate decision framework that mirrors OEM GT entry decisions, progressing from strategic intent to platform selection and final Go / No-Go recommendation.

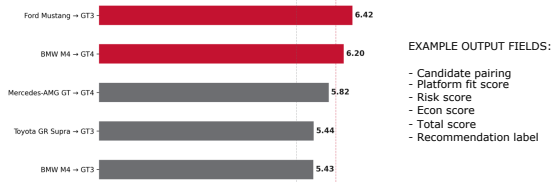
Data: We combine public motorsport benchmarks with synthetic datasets representing GT series constraints, conversion complexity, customer demand, and competitor economics.

Approach: We apply a multi-criteria scoring model with weighted KPIs, stage-gate filtering, and a simplified unit economics model to evaluate feasibility, risk, and profitability.



RESULTS

Our final Gate 5 output ranks vehicle-to-class candidates using **platform fit, risk, and economic feasibility**. The results show that top-ranked pairings tend to achieve strong platform scores but face non-trivial economic conversion costs, confirming that technical readiness and overall viability do not always align. Mid-tier candidates frequently underperform on a **single dimension** rather than across all three, suggesting that targeted mitigation can recover borderline cases. This structure gives the framework practical value beyond ranking by identifying where intervention is most likely to **shift a no-go toward a conditional go**.



EXAMPLE OUTPUT FIELDS:

- Candidate pairing
- Platform fit score
- Risk score
- Econ score
- Total score
- Recommendation label

Key Interpretation:

- Top-ranked pairings reveal a **tension between platform fit and economic feasibility**, confirming that technical readiness alone does not guarantee viability.
- Mid-tier candidates frequently **underperform on one dimension only**, meaning the ranking should be read as a diagnostic, not a verdict: a low economics or risk score may be addressable, while a low platform fit score is harder to overcome.
- The framework is most **valuable** when a sponsor already has a specific pairing in mind. It structures the conversation around what would need to change, not just whether entry is possible.

DISCUSSION



CONCLUSIONS

We developed a stage-gated analytics framework that reframes race-entry evaluation as a **tradeoff decision** rather than a technical compliance check.

The core contribution is the distinction between a platform that is **feasible** and one that is **advisable**, a gap the retail baseline comparison makes explicit for pairings such as Corvette to GT4.

Future work should incorporate real homologation documents and supplier-level cost data to sharpen the economic layer and move the tool closer to a full sponsor decision brief.